Example

Problem

❑ True or False?

- Q1. Insertion sort's ***worst-case*** running time is $O(n^2), \Omega(n^2)$, and $\Theta(n^2)$? True, True, True
  - ✓ The worst-case running time (n^2) grows **no faster than** the rate of growth of function n^2.
  - ✓ The worst-case running time (n^2) grows **at least** the rate of growth of function n^2.
  - ✓ The worst-case running time (n^2) grows **precisely** at the rate of growth of function n^2.

- Q2-1) Insertion sort's running time is $\Theta(n^2)$? False
  - ✓ Insertion sort's running time grows precisely at the rate of growth of function n^2.
- Q2-2) How about "Insertion sort's running time is $O(n^2)$"? True
  - ✓ Insertion sort's running time grows no faster than the rate of growth of function n^2.